

Information-optimal illumination for dead-time-limited single-pixel imaging: certificates, water filling, and the jitter-capped ridge

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Abstract

A photon-counting detector operated at its deterministic information-optimal flux can forfeit more than half of its recorded information the moment its dead time carries a few percent of timing jitter. This paper asks how hard a real, jittery detector can be driven, and with what illumination — one law answers the first half, one certified design principle the second, and a preregistered campaign seals both. The law: for hidden random hold times of coefficient of variation c_v , we derive an exact missing-information identity and the long-window count-information rate $J_\infty(\rho, c_v) = \rho/[(1+\rho)(1+c_v^2\rho^2)]$, whose optimum is finite for every nonzero hold-time variance, solves the cubic $c_v^2\rho^2(1+2\rho) = 1$, and scales as $2^{-1/3}c_v^{-2/3}$ at small jitter; matching this extensive loss to the deterministic terminal-phase loss of a companion study [1] predicts the smooth crossover $\rho^*(\nu, c_v) \sim (6\nu)^{1/3}(1+12\nu c_v^2)^{-1/3}$, with zero jitter the critical manifold on which the ridge grows as $\nu^{1/3}$ ($\nu = T/\tau$ dead-time slots per exposure). Exact-law peak predictions registered before the refined Monte-Carlo sweep completed were confirmed out-of-sample at both fresh points (3% and 6%), and a pooled two-run continuous-peak fit measures the jitter exponent at -0.658 (1.3% from $-2/3$). The design principle: hand-designed sparse and matched patterns are atoms of one concave experiment-design program whose per-atom information is the exact renewal kernel, and under uniform-dose safety that program yields a sharp doctrine. Unconstrained, information wants to concentrate load; a uniform $\pm 5\%$ per-pixel dose constraint removes that degree of freedom and collapses the tested target-load-normalized adaptive design toward a balanced pattern family; the near-optimality of that balanced design (SCAT32) over an expanded gain-coupled dictionary is certified, not assumed, by a full dose-constrained dual bound; and the surviving useful control is the global detector operating point, one power knob that tracks the jitter-capped ridge. A preregistered simulation campaign (M1) tests the doctrine with a paired fixed-dwell ridge operating-point primary (RIDGE-SCAT32 versus SCAT32-060, a power-for-time comparison), a

nine-dwell time-to-quality secondary, a 480-cell near-optimality certificate, five arms, and a mandatory disclosure ledger; the three verdicts are mutually non-rescuing and all outcomes are reported as frozen [M1: ridge operating-point, speed, and certificate verdicts with cross-arm numbers]. The result is an operating handbook: measure the jitter, let the law set the load, let the certificate vouch for the balanced patterns, and tune the global operating point when time is the scarce resource.

1 Introduction

Drive a photon-counting detector to the flux where its count information peaks, and a five-percent timing jitter in its dead time silently forfeits about 55% of that information (Table 1). Single-pixel imaging (SPI) and computational ghost imaging reconstruct a scene from the responses of a single bucket detector to structured illumination [2, 3, 4]; in the photon-starved regime the bucket is a photon-counting device that is blind for a dead time τ after each registered count, and the conventional response is to operate at low flux, keeping the per-slot load $\rho = \lambda\tau$ at the few-percent level and paying in acquisition time. A companion study [1] showed that this convention leaves gains on the table: the exact Fisher information of the integrated nonparalyzable count possesses a principal ridge at the cube-root load $\rho^*(\nu) = (6\nu)^{1/3} - 2/3 + O(\nu^{-1/3})$, and that work mapped when driving a fixed illumination family toward the ridge helps. The trap above is the sharp edge of that map: the deterministic ridge is not where a *real* detector should operate.

Two questions therefore remain open. First, a *deployment* question: how hard can a real detector — whose dead time is a random hold time with some measurable coefficient of variation — actually be driven before the ridge collapses? Second, a *design* question: under a fixed incident-photon and dose budget, which illumination extracts the most information from the dead-time channel, and by how much can a certified optimizer improve on a strong fixed hand design?

One law answers the first question: hidden hold-

time randomness adds an *extensive* missing-information term, the long-window optimum becomes finite at the root of an exact cubic, and the deterministic ridge is the zero-jitter critical manifold of a two-parameter crossover (Section 2); its peak predictions, registered before the refined Monte-Carlo sweep existed, were confirmed out-of-sample. One certified design principle answers the second: hand-designed patterns are atoms of a single concave experiment-design program over design measures with the exact renewal kernel, and under uniform-dose safety that program collapses toward a balanced pattern family whose near-optimality is certified against an expanded frozen dictionary, leaving the global detector operating point as the surviving useful control (Section 3). A preregistered campaign (M1) seals both with a ridge operating-point primary, a time-to-quality secondary, a near-optimality certificate, and a mandatory disclosure ledger (Sections 4–5).

We formulate a local approximate-design framework for single-pixel illumination in which each pattern’s information is computed from the exact finite-window nonparalyzable renewal count law, optimized under explicit detector-load, dose, per-pixel-exposure, and peak-irradiance constraints and accompanied by generalized-equivalence and exact-rounding certificates. Existing photon-counting optimal experiment design, adaptive or learned SPI, and dead-time optimal-flux studies address neighboring components [5, 6, 7, 8, 9, 10]; we found no prior framework combining these elements for integrated-count single-pixel illumination. The certificate is always stated relative to the frozen dictionary, the local pre-scan estimate, and the declared resource model.

This work contributes: (i) the random-hold count-information law (Section 2): an exact missing-information identity, the long-window rate with its exact jitter-limited cubic optimum and small-jitter coefficient $2^{-1/3}$, and the matched crossover prediction $(6\nu)^{1/3}(1 + 12\nu c_v^2)^{-1/3}$, verified out-of-sample against preregistered predictions with a pooled jitter-exponent measurement of -0.658 ; (ii) the certified design principle (Section 3): one concave program with the exact renewal kernel whose uniform-dose-constrained optimum collapses to a balanced pattern family, a full dose-constrained dual near-optimality certificate over an expanded gain-coupled dictionary, frozen guards, and the photon-budget versus time-budget conjugacy as its operating doctrine; (iii) a preregistered simulation campaign (M1) whose scenes, atoms, load grid, guards, endpoints, and three mutually non-rescuing verdicts were frozen in an immutable commit before any confirmatory reconstruction existed, reported as frozen including any negative primary [M1: campaign outcomes]; and (iv) an operating handbook for real detectors (Table 5), ending in a bench-testable prediction. All code, frozen prereg-

istration documents, and per-cell evidence bundles are openly released; full derivations, guard machinery, and verification detail are in the supplement.

2 The law: driving a jittery detector

The recovery-time law of the detector determines a scalar count-information kernel; everything else in this paper allocates illumination against that kernel. The companion’s deterministic detector [1] is the zero-jitter critical manifold of the theory below. Proofs at manuscript rigor are given in the supplement.

2.1 Random-hold count information

Generalize the detector so that after each registered event it blocks for a hidden *random* hold time $B_i \sim H$, iid, independent of the Poisson arrivals and of λ , with mean τ and coefficient of variation $c_v = \sigma_B/\tau$ (the companion’s deterministic model is $c_v = 0$). The observed datum remains the integrated registered count N_T over $T = \nu\tau$; the active time L_T , the state path, and the realized holds are hidden from the count-only observer.

Theorem 1 (missing-information identity) *For $\theta = \log \lambda$, the complete detector-path information is $\mathbb{E}[N_T]$, and the information retained by the scalar count is exactly*

$$I_N(\theta; T) = \mathbb{E}[N_T] - \lambda^2 \mathbb{E}[\text{Var}(L_T | N_T)] \quad (1)$$

for every finite T and every iid hidden hold law H independent of λ .

The companion’s terminal-phase identity is the special case in which the uncertainty in L_T is confined to the terminal detector phase. The identity requires amendment if the hold law depends on λ or the scene, if realized holds are observed and used, if holds are serially correlated, or if the counter is paralyzable.

Theorem 2 (long-window rate) *Under standard nonlattice renewal local-limit conditions,*

$$\frac{I_N(\log \lambda; T)}{\nu} \rightarrow J_\infty(\rho, c_v) = \frac{\rho}{(1 + \rho)(1 + c_v^2 \rho^2)}, \quad (2)$$

depending on H only through its mean and variance. The missing-information rate converges to $c_v^2 \rho^3 / [(1 + \rho)(1 + c_v^2 \rho^2)]$: an extensive loss equal, to first order in $c_v^2 \rho^2$, to the heuristic $\lambda^2 \sigma_B^2 \mathbb{E}[N_T]$ — conditioning on the count partially reveals the accumulated hold fluctuation, contributing the factor $(1 + c_v^2 \rho^2)^{-1}$ — and, unlike the deterministic terminal-phase loss, nonvanishing as $\nu \rightarrow \infty$.

Theorem 3 (exact jitter-limited optimum) For every fixed $c_v > 0$, $J_\infty(\rho, c_v)$ has a unique interior maximum ρ_c solving the cubic

$$c_v^2 \rho_c^2 (1 + 2\rho_c) = 1, \quad (3)$$

with small-jitter expansion

$$\rho_c = 2^{-1/3} c_v^{-2/3} - \frac{1}{6} + \frac{2^{1/3}}{36} c_v^{2/3} + O(c_v^{4/3}), \quad (4)$$

whose leading coefficient is $2^{-1/3} = 0.79370\dots$

For exponential holds ($c_v = 1$, outside the small-jitter regime) the exact positive root is $\rho_c = 0.6573\dots$; gamma, bounded two-point, and lognormal holds sharing a small c_v share the leading coefficient, with distribution shape entering at the next order.

2.2 The crossover and its universality class

Matching the extensive loss $c_v^2 \rho^2$ against the deterministic terminal-boundary loss $\rho^2/(12\nu)$ in the joint regime $\nu \rightarrow \infty$, $c_v \rightarrow 0$, $c_v^2 \nu = O(1)$ gives

$$\rho_*(\nu, c_v) \sim \left(2c_v^2 + \frac{1}{6\nu}\right)^{-1/3} = (6\nu)^{1/3} (1 + 12\nu c_v^2)^{-1/3}, \quad (5)$$

with scaling variable $x = 12\nu c_v^2$ and crossover at $c_v \asymp (12\nu)^{-1/2}$: the deterministic limit $x \rightarrow 0$ recovers $(6\nu)^{1/3}$ and the jitter-dominated limit recovers $2^{-1/3} c_v^{-2/3}$. An earlier $\min\{(6\nu)^{1/3}, c_0 c_v^{-2/3}\}$ conjecture is only the two-limit envelope of Eq. (5), with an artificial kink, and is superseded. Per the frozen claim discipline, the exact identity and the long-window limit *imply* the jitter exponent $-2/3$; the full finite-window crossover of Eq. (5) is a matched-asymptotic *prediction* until a uniform two-parameter (triangular-array renewal local-limit) proof is complete [supplement: uniform crossover proof, or the prediction label is retained].

The exponent pair $(1/3, -2/3)$ is universal within the regular finite-variance iid hidden-hold class: iid hidden nonnegative holds whose law is independent of λ , with finite nonzero mean and finite variance, forming a regular small-jitter family, under count-only observation and active-start nonparalyzable operation. It is *not* claimed for heavy-tailed (infinite-variance) holds, correlated or drifting holds, holds observed by the electronics, paralyzable detectors, event-time observations, or λ -dependent recovery. The precise critical statement is: zero hold-time variance is the critical manifold on which the optimal load continues to grow with window length; any fixed nonzero hidden hold variance produces a finite long-window optimum. Adjacent renewal-CLT and random-dead-time literature establishes the mean/variance ingredients and task-specific finite-rate optima [11, 12, 13, 14, 10]; we found no source stating the count-only exponent pair or the extensive/terminal crossover.

Table 1: Scalar count information $J(\rho, \nu = 2000, c_v)$ for lognormal hold times: three-stage peak measurement against the exact-law predictions of Theorem 3/Eq. (5), registered *before* the data (docs/R14_PREREGISTERED_PREDICTIONS.md, commit 8a627bb; verdicts at 1d8d7aa, 4d8310d, bd731d5). Grid: jitter_sfi_v2, 100,000 frames, 25-point log grid (spacing 1.157). Zoom: continuous quadratic-interpolated peaks, local run, 150,000 frames. Replication: independent Colab run, 400,000 frames. Both out-of-sample (o.o.s.) grid rows hit ($c_v = 0.02$: 3%; $c_v = 0.2$: 6%); the pooled eight-measurement log-log fit gives slope -0.658 vs. $-2/3$ (1.3%), with the constant consistent with $2^{-1/3} = 0.794$ within 5–10% (the earlier coarse-grid fit 0.72 was a finite-grid artifact) and a reproducible mild $\approx -8\%$ low bias at the end points (open higher-order note). Hardware warning: at the deterministic ridge ($\rho = 22.3$), 5% jitter leaves $J = 0.43$ vs. 0.95 — about 55% of the scalar count information forfeited. J at peak from the grid run.

c_v	peak ρ (grid / zoom / replic.)	exact law	J at peak
0	22.297 / — / —	22.25 (ridge)	0.9452
0.02	10.739 / 9.645 / 9.610	≈ 10.4 (o.o.s.)	0.8731
0.05	5.173 / 5.154 / 5.700	5.69	0.7848
0.1	3.862 / 3.752 / 3.399	3.50	0.6989
0.2	2.153 / 2.106 / 2.051	≈ 2.30 (o.o.s.)	0.5882
0.3	1.607 / — / —	1.63	0.5042

2.3 Preregistered verification

The law was tested the way the campaigns are run: predictions first, data second. The exact-law column of Table 1, including two out-of-sample rows, was registered before any refined measurement existed (commit 8a627bb). The grid sweep then confirmed the law at both registered fresh points — $c_v = 0.02$: predicted ≈ 10.4 , measured 10.739 (3%); $c_v = 0.2$: predicted ≈ 2.30 , measured 2.153 (6%) — with every in-sample peak within one grid notch and the $\nu = 200$ column consistent including the crossover correction of Eq. (5). A continuous-peak zoom and an independent replication (Colab, 400,000 frames) then measured the exponent itself: the pooled eight-measurement log-log fit gives slope -0.658 , within 1.3% of the predicted $-2/3$, with the constant consistent with $2^{-1/3}$ within 5–10% and a reproducible mild $\approx -8\%$ end-point low bias left as an open higher-order note (candidate: a lognormal third-moment correction). One honest beat from the record: a mid-sweep effective-multiplier hypothesis ($k = 4/3$) was preregistered and refuted by the very next measured point (commits 5e51088, 1a32758). Run specifications, both-run peak tables, and the frozen three-family exponent battery (tests E1–E5) that remains a preregistered extension are given in the supplement.

3 The certified design principle

The design principle is a doctrine in four movements. Unconstrained, the information geometry wants to concentrate illumination load on the brightest, most valuable directions (Section 3.1). A uniform per-pixel dose constraint removes that degree of freedom: within the frozen target-load-normalized atom family the adaptive concentration collapses toward the balanced design (Section 3.2). The simplicity of that balanced design is then certified rather than assumed, by a full dose-constrained dual bound over an expanded gain-coupled dictionary (Section 3.3). What survives as useful control is not spatial redistribution of dose but the global detector operating point, a single power knob that tracks the jitter-capped ridge (Sections 3.4 and 3.6).

3.1 One concave program over design measures

Freeze the coarse pre-scan reconstruction at $\hat{x}$ and parameterize the locally estimable image perturbation as $x(\theta) = \hat{x} + B\theta$, with $B \in \mathbb{R}^{n \times r}$ spanning a declared r -dimensional estimable subspace. For a nonnegative illumination atom a the arrival rate is $\lambda(a) = \Phi a^\top \hat{x} + d$ and the load is $\rho(a) = \tau\lambda(a)$. With $J_{\text{exact}}(\rho, \nu)$ the exact per-slot information of the integrated renewal count for $\log \lambda$ (generalized to the random-hold kernel of Theorem 2 when $c_v > 0$) and score direction $q(a) = \Phi B^\top a / \lambda(a)$, the rank-one local information atom is the chain-rule form

$$\begin{aligned} H(a; \hat{x}) &= \nu J_{\text{exact}}(\rho(a), \nu) q(a) q(a)^\top \\ &= \nu J_{\text{exact}}(\rho(a), \nu) \frac{\Phi^2}{\lambda(a)^2} B^\top a a^\top B. \end{aligned} \quad (6)$$

A raw sensitivity $g \propto \Phi a$ is incomplete unless the factor $\sqrt{\nu J_{\text{exact}}}/\lambda$ is absorbed into it. For a probability measure ξ over the compact feasible atom set $\mathcal{A}$, define $V(\xi) = V_0 + \int_{\mathcal{A}} H(a) d\xi(a)$ with $V_0 \succeq 0$ the frozen pre-scan information, and maximize the local D-criterion

$$\max_{\xi} \log \det V(\xi) \quad \text{s.t. linear resource constraints,} \quad (7)$$

subject to average load, cumulative dose, family allocation, and peak constraints. The objective is concave in ξ ; it is a *local information surrogate* and is not claimed to optimize reconstruction PSNR or any image-domain risk. The estimable subspace is frozen at $r = 200$ (the top eigenbasis of the fixed comparator’s information matrix, fixed before the solve, with a fixed numerical ridge; conventions in the supplement). If every atom obeys a pointwise equal-load constraint the scalar kernel factors out and the geometry reduces to ordinary linear D-optimality [15]; genuine dead-time-aware design therefore requires controlled load variation. Unconstrained, this geometry rewards concentrating load

on the brightest, highest-value directions — the incentive whose fate under safety constraints the rest of this section traces.

3.2 Uniform-dose safety and the collapse of adaptive concentration

The unconstrained incentive to concentrate load does not survive the safety constraints. The earlier variable-load adaptive arm (OED-DT) is therefore retired as a proposed or gated imaging method: its speed gate was withdrawn before the campaign freeze as infeasible, not recorded as a confirmatory failure. Within the frozen target-load-normalized atom family, under the incident-signal budget and the $\pm 5\%$ per-pixel dose constraint, no nonzero adaptive mixture is admissible in the audited development cell — the descending exact mixture scan terminates at zero adaptive rows, `ADAPTIVE_COLLAPSE_UNDER_GUARDS`. This is development evidence about one parameterization, explicitly not a confirmatory theorem and not a claim that adaptivity is foreclosed: the target-load-normalized atomization fixes each pattern’s amplitude to hit a predicted load, which on dim supports demands large physical amplitude and anti-aligns the dose vector with brightness, so the collapse could be an artifact of that anti-brightness parameterization. Removing that possibility is exactly why the near-optimality certificate below is run over an *expanded* gain-coupled class. The descending scan survives only as a path-specific descriptive diagnostic (`PATH_FEASIBLE_ALPHA`, relative to the frozen row ordering, reported in the methods-development supplement), never an estimator, fallback, or certificate. The dose-relaxed and dose-constrained continuous OED solutions are retained as no-endpoint design diagnostics that illustrate the incentive and the safety frontier (Figure 1); neither is the proposed imaging method.

3.3 Simplicity certified, not assumed

The balanced design’s near-optimality is certified, not trusted. By concavity, optimality is equivalent to a bounded reduced sensitivity (the constrained Kiefer–Wolfowitz / general equivalence condition for nonlinear models [15, 16]). The confirmatory quantity is the full dose-constrained dual upper bound G_{full} of the deployed balanced SCAT32 design against the continuous optimum of the dose-safe class, evaluated over an *expanded* dictionary $D_{\text{cert}} = D_{\text{load}} \cup D_{\text{gain}}$: to the frozen target-load-normalized atoms D_{load} it adds a certificate-only gain-coupled family D_{gain} in which each base geometry is scaled by $\gamma \in \{0.2, 0.5, 1, 2, 5\}$ with its predicted load allowed to *emerge* rather than be rescaled to a target, so fixed-flux self-water-filling and bounded per-pattern gain control lie inside the comparison class. The dual carries one incident-budget multi-

Table 2: The frozen design machinery at a glance (R10/R11/R13/R17; full statements in the supplement).

Element	Frozen rule
Dictionary	all cyclic translates of the 13 R10 support families (scattered $k = 16/32$, solid 4×4 , Lblob, bars, compact-32, ring); weight powers $p \in \{0, 1\}$; the certificate scan adds gain-coupled copies $a_{g,\gamma} = \gamma a_g^{\text{base}}$, $\gamma \in \{0.2, 0.5, 1, 2, 5\}$, emergent load ($D_{\text{cert}} = D_{\text{load}} \cup D_{\text{gain}}$)
Budget	incident-signal inequality $\int \rho_s d\xi \leq 0.60$; detected counts dual-reported, never the sole budget
Peak	shape: $w_j \in [\frac{1}{4}, 4]$, $k \geq 16$; physical: $\max a_{ij} \leq 1536$
Dose	per-pixel mean exposure within $\pm 5\%$ of image mean, re-verified post-rounding
Floors	$V_{\text{OED}} \succeq 0.5 V_{\text{FIXED}^*}$; $\text{tr}(V^{-1}) \leq 1.05 \text{tr}(V_{\text{FIXED}^*}^{-1})$ (disclosed separately; not in the dual target class)
Kernel trust	exact-PMF certification; $J_{\text{mean}}/J_{\text{exact}} \geq 0.90$; RQL log-load bias ≤ 0.05 ; $p_{\text{ceil}} \leq 0.05$ per atom, ≤ 0.01 design-weighted
Certificate	$G_{\text{full}}/r \leq 10^{-2}$ (full dose-constrained dual, expanded D_{cert} scan; local D-eff. ≥ 0.99005)
Rounding	exact 972-row design, $\text{Eff}_D \geq 0.95$

plier and every upper and lower per-pixel-dose multiplier, uses coverage-seeded column generation, and finishes with an exact scan over all of D_{cert} ; the frozen bar is $G_{\text{full}}/r \leq 10^{-2}$, certifying local D-efficiency of at least $\exp(-0.01) = 0.99005$ relative to the continuous optimum of the expanded dose-safe class (full dual in the supplement). A-risk and spectral constraints are deliberately excluded from the dual target class, so that the certified class is larger and the near-optimality statement stronger; the deployed SCAT32 rows still satisfy every physical and conditioning disclosure separately. The frozen certificate wording is exact. Within the frozen nonnegative illumination dictionary, local pre-scan linearization, incident budget, and $\pm 5\%$ per-pixel dose constraint, the balanced SCAT32 design was tested against a full constrained dual upper bound. Global ridge tracking was evaluated separately as the time-limited, power-available control. Continuous weights are rounded to an exact 972-row design by efficient rounding and deterministic exchange [17, 18, 19], admissible only at exact D-efficiency ≥ 0.95 . Every remaining guard is a frozen constraint, not a tunable proxy (Table 2); the full guard mathematics, the expanded dictionary listing, and the budget accounting are in the supplement.

3.4 The conjugacy doctrine: photon budget versus time budget

The doctrine has two conjugate resource corners of the same dead-time information map. In the *photon-budgeted* corner the near-optimality certificate holds the incident-signal resource as a hard *inequality*,

$$\int \rho_s(a) d\xi(a) \leq 0.60, \quad (8)$$

so a design measure may push a subset of high-value atoms into the ridge zone only if compensated by low-load or omitted atoms (R11 §4); detected counts are dual-reported but are never the sole fairness budget, because $\mathbb{E}[N | \rho] \simeq \nu\rho/(1 + \rho)$ has vanishing marginal cost at high load. The *time-budgeted* corner is the no-gate ridge-tracking policy RIDGE-SCAT32: it fixes the balanced support and applies one global multiplier per dwell, calibrated on the pre-scan estimate to mean load $\rho_R(\nu)$, followed by a single global safety clip, spending a deliberately larger incident budget for time rather than photons (R11 §2). The exact production ridge is $\rho_R(\nu) = \min \arg \max_{\rho \in \mathcal{R}_{\text{adm}}(\nu)} J_{\text{exact}}(\rho, \nu)$ over the admissible set $\mathcal{R}_{\text{adm}}(\nu) = \{0 < \rho \leq 24 : p_{\text{ceil}} \leq 0.01, J_{\text{mean}}/J_{\text{exact}} \geq 0.90\}$, using the exact numerical ridge rather than the asymptotic cube-root form (R11 §1). Under the incident-photon budget the relevant scalar efficiency is $J_{\text{exact}}(\rho, \nu)/\rho$, which decreases across the admitted palette; a photon-budgeted optimizer therefore rationally spends on lower loads and *avoids* the ridge, while the ridge belongs to the time-limited, power-available corner. This is a resource-regime statement, not a universal prohibition on ridge-zone allocation: sufficiently valuable directions may still receive larger loads in other cells (R13 §8(f)).

3.5 One program behind every optimum

Every true interior optimum of the declared design problem — over load, support scale, and intensity weighting — satisfies one generalized reduced-sensitivity/KKT system of the master program, whose load coordinate is the information-water-filling equation and whose scale and weighting coordinates price conditioning through the V^{-1} geometry (full system in the supplement); this also explains why raising proxy contrast from $p = 1$ to $p = 2$ can lower the true objective. One caveat is kept deliberately: the master design problem places load, support scale, and intensity weighting in one constrained concave measure optimization, and its support atoms satisfy a common reduced-sensitivity condition; but the empirically observed occupancy ($k \approx 32$) and weighting ($p \approx 1$) optima *motivate* these coordinates and are not themselves claimed as continuous KKT theorems — they were selected from

discrete grids and judged by image endpoints, not by the master log det objective.

3.6 Why one global knob suffices for RIDGE-SCAT32

Fixed flux performs implicit information water filling when bucket brightness and local directional value satisfy the marginal-information alignment condition $\ell_i J'(\kappa u_i) = \beta$ across active directions ($\ell_i = q_i^\top V^{-1} q_i$ the directional value, $u_i = a_i^\top x$ the bucket brightness); its adequacy is measured by a projected KKT residual — not C_u alone — and outside that condition, variable-load OED can improve without any universal bound on the fixed-flux approximation ratio. On non-negative sparse-support scenes the source-induced load ordering naturally tracks directional value, which is why RIDGE-SCAT32 needs only a single global multiplier, and why the uniform per-pattern servo — which loses by up to ~ 7 dB in the companion’s dev evidence [1] — fails by erasing exactly that alignment. With random holds, J' vanishes at the jitter cap and is negative beyond it, so fixed flux can overdrive the brightest directions and explicit gain control becomes necessary. The full proposition, its residual near-optimality bound, and its failure modes are in the supplement. Naive per-pattern load equalization was thus refuted (R11) for destroying this brightness–information alignment, while the dose-safe analysis of Section 3.2 shows the opposite extreme — adaptive concentration — conflicting with strict dose and conditioning safety; the balanced global-flux policy is the robust middle.

4 Preregistered campaign (M1)

M1 is a separate follow-up campaign with its own seed namespace, manifests, and expected-cell ledger; it does not reopen the companion’s frozen studies. It freezes on the R13 audit plus one immutable commit (tag `m1-freeze`); after that commit no scene, atom family, load grid, guard, endpoint, or gate may change, and if the primary fails there is no third redesign.

Geometry and constants. $n = 32^2 = 1024$; $M_{\text{total}} = 1024 = M_{\text{pre}}(52) + M_{\text{main}}(972)$ for every arm; nonparalyzable active start, $\tau = 50$ ns, $d = 0$, $\sigma_b = 0$; $\bar{\rho} \in \{0.05, 0.60\}$ for the budget-anchored arms; $\nu \in \{5, 10, 20, 50, 100, 200, 500, 1000, 2000\}$; five measurement seeds per (image, condition). Optical time $T_{\text{opt}} = M_{\text{total}} \nu \tau$; design-computation wall time is reported separately and never folded into T_{opt} .

Scenes. Twenty-four fresh confirmatory instances (six generator families $\times$ four instances) with seeds $s_{f,r} = 633000 + 100f + r$, plus six development-only

instances $632900 + f$. The six committed generators (glyph, chirp, spokes, maze, contour, microtexture) are reused verbatim with their actual names — “USAF” is not used unless a USAF target is generated, and spokes is described as a radial resolution target. No acceptance filter of any kind is applied; generators, images, and ROIs receive SHA256 manifests.

Common pre-scan. Every arm receives and pays for the same 52-pattern scene-independent balanced pre-scan at its own $(\bar{\rho}, \nu)$: 32 paired 4×4 -block rows (support 32, amplitude 32), 16 block rows (amplitude 16), and 4 quadrant rows (amplitude 4), so every pixel receives cumulative amplitude $32 + 16 + 4 = 52$ and the row-mean pattern is exactly the all-ones vector; the mean pre-scan load then equals the declared global load for any sum-normalized scene, with no $\hat{x}$ -dependent row scaling (R13 §2). All 52 counts enter every arm’s final reconstruction; RIDGE-SCAT32 additionally uses the pre-scan estimate to calibrate its single global source multiplier, with no per-pattern servo. Every arm runs the same 972 SCAT32 main patterns and totals exactly 1024 physical exposures.

Arms and the deployed design. The deployed spatial design is the balanced exact 972-row SCAT32 multiset, identical across every operating-point comparison; the common 52-row pre-scan is charged to each arm. Table 3 lists the five arms. Three share this design and differ only in source power: SCAT32-SAFE at global mean load 0.05, SCAT32-060 at 0.60, and RIDGE-SCAT32, which applies one global multiplier per dwell (calibrated on the pre-scan to $\rho_R(\nu)$, then the frozen global clip). Two frozen fixed baselines, SCAT16 and LBLOB16 at load 0.60, are retained as descriptive context columns (no gate), so the cross-arm occupancy-ladder comparison keeps its reference columns. The $k = 32$ occupancy rung of the deployed design is fixed by the frozen development rule (median fixed-dwell radiometric PSNR at $(\bar{\rho}, \nu) = (0.60, 2000)$, averaged over development seeds), which selected SCAT32 over the clustered LBLOB16 and a matched baseline with development scores 17.498, 11.908, and 9.181 dB respectively; an earlier information-score-based selection was invalidated by external audit (R15) for departing from the frozen rule and was re-run under the original R10 prescription. The rung is consistent with the companion’s Study 2 fixed-dwell optimum [1]. Reconstruction (RQL with the frozen analytic λ_{TV} rule and pooled initialization) is identical across arms.

Endpoints and gates. Three preregistered, mutually non-rescuing verdicts govern the campaign. The *primary* is a paired fixed-dwell ridge operating-point test (RIDGE.OPERATING.PASS). At terminal dwell

Table 3: The five M1 arms over the common balanced SCAT32 design. The paired RIDGE-SCAT32 vs. SCAT32-060 comparison carries the primary gate and RIDGE-SCAT32 vs. SCAT32-SAFE the speed gate; the near-optimality certificate (DOSE_SAFE_CERT_PASS) applies to the deployed design itself. SCAT16 and LBLOB16 are descriptive context.

Arm	Role	Gate
SCAT32-SAFE	balanced design at load 0.05 (safe reference; speed baseline)	speed base
SCAT32-060	balanced design at load 0.60 (primary comparator)	primary comp.
RIDGE-SCAT32	one global multiplier per dwell to $\rho_R(\nu)$ + global clip (ridge policy)	primary, speed
SCAT16	frozen scattered $k = 16$ at load 0.60 (companion bridge)	none
LBLOB16	committed clustered support at load 0.60 (spectral-floor reference)	none

$\nu = 2000$, for each confirmatory image j define $\Delta Q_j^{\text{ridge}} = \overline{\text{PSNR}}_{\text{rad}}(\text{RIDGE-SCAT32}, \nu=2000, j) - \overline{\text{PSNR}}_{\text{rad}}(\text{SCAT32-060}, \nu=2000, j)$, the mean over the five frozen seeds, with identical pre-scan, identical 972-row SCAT32 multiset, paired pattern/noise seeds, the same reconstruction path, and the same optical dwell — only the single global main-acquisition multiplier differs. PASS iff $\text{median}_j \Delta Q_j^{\text{ridge}} \geq 1.0$ dB, the 2.5% bootstrap lower bound on the median exceeds 0, and $\#\{j : \Delta Q_j^{\text{ridge}} > 0\} \geq 18/24$, using the existing 10,000-replicate nested family-stratified bootstrap; any numerical or guard failure stays in the intention-to-treat cohort and counts as nonpositive. This is a fixed-dwell, power-for-time operating-point test. It is not an equal-photon-budget or photon-efficiency comparison. The *secondary* is the nine-dwell Q_{90} time-to-quality test (RIDGE.SPEED.PASS): SCAT32-SAFE at load 0.05 against the dwell-dependent RIDGE-SCAT32 policy, using the frozen PAVA/censoring machinery (the seed-mean radiometric PSNR, equal-weight PAVA isotonic curves, 0.50 dB safe-range informativeness floor, and censoring taxonomy transfer verbatim), with median $S_{\text{gate}} \geq 3$, bootstrap lower bound > 1 , and $\geq 18/24$ images with $S_{\text{gate}} > 1$. The *confirmatory structural secondary* is the near-optimality certificate (DOSE_SAFE_CERT_PASS, Section 3.3): PASS iff every one of the $24 \times 5 \times 2 \times 2 = 480$ certificate cells ($\nu \in \{200, 2000\}$, $b \in \{0.05, 0.60\}$) is primal-feasible, numerically certified, and satisfies $G_{\text{full}}/r \leq 10^{-2}$. No verdict can rescue another.

Disclosure ledger. Every cell discloses incident and detected photons (predicted and realized), the incident-dose and detected-count ratios, source/peak-irradiance ratio, the achieved mean load and its quantiles ρ_5 , ρ_{50} , ρ_{95} , ρ_{max} , the physical peak, predicted and realized ceiling fractions, any RIDGE_GUARD_CLIPPED flag, and mean J_{exact} ; certificate cells add the full dose-constrained dual G_{full} , the exact D-efficiency, the dual/complementarity report, and the MU_CAP_ACTIVE disclosure. The full nine-dwell kernel/ridge certification table and the SHA256 of $D_{\text{cert}} = D_{\text{load}} \cup D_{\text{gain}}$ are frozen before any confirmatory scene is opened.

5 Results

5.1 Primary: ridge operating-point gain

[M1: RIDGE_OPERATING_PASS — median ΔQ^{ridge} (dB) at $\nu = 2000$, 2.5% bootstrap lower bound, count of scenes with $\Delta Q^{\text{ridge}} > 0$, and the frozen PASS/FAIL call, with the mandatory incident-dose ratio, detected-count ratio, achieved mean load and quantiles, physical peak, ceiling fraction, and RIDGE_GUARD_CLIPPED tally.] This is a fixed-dwell, power-for-time operating-point test. It is not an equal-photon-budget or photon-efficiency comparison.

5.2 Secondary: ridge time-to-quality

[M1: RIDGE_SPEED_PASS — median S_{gate} , bootstrap lower bound, count of scenes with $S_{\text{gate}} > 1$, and PASS/FAIL, for SCAT32-SAFE against the dwell-dependent RIDGE-SCAT32 policy.]

5.3 Confirmatory certificate: near-optimality of the balanced design

[M1: DOSE_SAFE_CERT_PASS — distribution of G_{full}/r across the 480 certificate cells, the fraction certified at $\leq 10^{-2}$, primal feasibility, and the frozen PASS/FAIL call.]

5.4 Context arms and design diagnostics

[M1: SCAT16/LBLOB16 context ladder, the dose-relaxed and dose-constrained OED design diagnostics feeding Figure 1, and the development-cell guard-versus- α collapse — summary here; full tables in the supplement.]

[M1 (Act III, four panels): (a) dose deviation, A-risk, and spectral guard versus path-specific α on a DEV representative, labeled development evidence; (b) confirmatory G_{full}/r distributions over the expanded dose-safe class (480 cells); (c) ridge-versus-0.60 fixed-dwell reconstructions and per-scene quality gains at $\nu = 2000$; (d) a schematic distinguishing the photon-budgeted and time-limited resource corners.]

Figure 1: Act III: uniform-dose safety collapses adaptive concentration (development evidence), the balanced design’s near-optimality is certified over the expanded class, and global operating-point control survives. Panels fill from the frozen M1 analyzer after tag **m1-freeze**.

Table 4: Cross-arm summary (descriptive; the gates are the RIDGE-SCAT32 primary and speed comparisons and the deployed-design certificate). All entries fill from the frozen M1 analyzer after tag **m1-freeze**; “—” marks a paired metric that does not apply to a reference or context arm.

Arm	Role	achieved $\bar{\rho}$	inc.-dose ratio	ceiling frac.	mean J_{exact}	$\Delta Q^{\text{ridge}}/S_{\text{gate}}$
SCAT32-SAFE	safe ref. / speed base	[M1]	[M1]	[M1]	[M1]	—
SCAT32-060	primary comparator	[M1]	[M1]	[M1]	[M1]	—
RIDGE-SCAT32	ridge policy (gated)	[M1]	[M1]	[M1]	[M1]	[M1]
SCAT16	context	[M1]	[M1]	[M1]	[M1]	—
LBLOB16	context	[M1]	[M1]	[M1]	[M1]	—

6 Discussion: the operating handbook

The campaign exposes two conjugate resource regimes of the same dead-time information map. Under a linear incident-photon budget the per-photon efficiency $J_{\text{exact}}(\rho, \nu)/\rho$ falls monotonically, so a budget-constrained design rations photons onto lower loads and *avoids* the ridge; the ridge belongs to time-constrained operation, where power is available and the only scarcity is dwell. Reading the flux ridge and the design allocation as one program clarifies that the large fixed-dwell gains of the companion are an operating-point effect, not evidence that a design must spatially redistribute dose into the ridge.

Contingent on the frozen verdicts ([M1: **RIDGE_OPERATING_PASS**, **RIDGE_SPEED_PASS**, and **DOSE_SAFE_CERT_PASS** calls]), the campaign is preregistered to license the governing ruling’s frozen conclusions. Under the declared dose-safety constraints, per-pattern load adaptation provided no certified material D-information advantage over the balanced design at the preregistered anchor dwells. The useful surviving control was the global operating point rather than spatial redistribution of incident dose. This does not assert that adaptivity is universally foreclosed or that the balanced design is optimal over all physical patterns: the collapse is a structural statement about the tested target-load-normalized family under strict uniform-dose safety, and the certificate

is stated relative to the frozen expanded dictionary, the local pre-scan estimate, and the declared resource model. The line’s thesis, in the frozen wording of the governing ruling, is this. Strict uniform-dose safety collapses the tested adaptive design space toward a balanced pattern family, for which near-optimality can be certified; high-flux benefit is then recovered by tuning the global detector operating point.

The jitter-capped ridge is the practical caution. The cube-root ridge is a property of the zero-jitter critical manifold; the moment a detector’s hold time carries a coefficient of variation c_v , the long-window optimum is finite — the exact cubic $c_v^2 \rho^2 (1 + 2\rho) = 1$ of Theorem 3, scaling as $2^{-1/3} c_v^{-2/3}$ — and driving past it is actively harmful: at the deterministic ridge ($\rho = 22.3$), 5% jitter already forfeits about 55% of the scalar count information in the refined sweep (Table 1). A deployment must therefore locate its operating point against the jitter-limited ceiling of the real detector, not the deterministic ridge; the safe conclusion is conditional on measured timing jitter.

Together, the law and the design principle compress into the operating handbook of Table 5: characterize the detector, let the law set the load ceiling, check engagement with the pre-scan, and pick the pattern recipe by which resource is scarce.

Two scope boundaries stand. All results are simulation-based, and every certificate is stated relative to the frozen dictionary, the local pre-scan estimate, and the declared resource model; an optical extension

Table 5: Operating handbook for dead-time-limited single-pixel imaging. The engagement index Γ is descriptive (the companion’s computable engagement criterion [1]), never a confirmatory gate.

Step	Rule
1. Characterize	measure τ and the hold-time c_v from time-tagged recovery intervals
2. Set the load	jitter-capped optimum from $c_v^2 \rho^2 (1 + 2\rho) = 1$ (long window) or $(6\nu)^{1/3} (1 + 12\nu c_v^2)^{-1/3}$ (finite window); never operate a jittered detector at the deterministic ridge
3. Check engagement	common 52-pattern pre-scan $\rightarrow$ measured C_u ; $\Gamma = C_u \sqrt{\nu \rho / (1 + \rho)} \gtrsim 1$ marks the photon-limited regime
4. Pick patterns	photon-budgeted: the balanced SCAT32 design, certified near-optimal over the expanded dose-safe class ($G_{\text{full}}/r \leq 10^{-2}$); time-budgeted: RIDGE-SCAT32 — one global multiplier to $\rho_R(\nu)$ (Section 3.6)
5. Doctrine	photon-constrained $\rightarrow$ stay below the ridge (J_{exact}/ρ falls); time-constrained $\rightarrow$ track the jitter-capped ridge

along the companion’s discussion — a coarse complex-field estimate synthesizing an optical displacement that removes common-mode photons before dead-time loss — remains a natural bench direction [1].

The law itself, however, is bench-testable now, with no imaging required. For a real single-photon counting module, the hold-time coefficient of variation is directly measurable from time-tagged recovery intervals; the theory then predicts a count-information peak at the root of $c_v^2 \rho^2 (1 + 2\rho) = 1$ rather than at the deterministic ridge — for $c_v = 0.05$, a peak near $\rho \approx 5.7$ instead of ≈ 22 at $\nu = 2000$, with about half the information forfeited if the deterministic ridge is used instead (Table 1). Sweeping source power at fixed dwell and estimating the count information from repeated frames traces $J(\rho)$ directly; this single-detector power sweep is the immediate bench experiment for the line.

7 Reproducibility

The design formulation, guards, and campaign follow five frozen rulings: the dead-time OED formulation and equivalence certificate (R10), the ridge-zone load architecture and incident-budget conjugacy (R11), the M1 freeze audit (R13), the unification ruling (R14, issue “R14 ruling: unification architecture”), whose theorem hierarchy Section 2 follows, and the safety-constrained refreeze (R17, issue “R17 ruling: safety-constrained OED collapse and coherent refreeze”), which retired the adaptive imaging arm, rearchitected the campaign around the ridge operating-point primary and the expanded-class near-optimality certificate, and froze the manuscript claim discipline

of Section 3. Exact-law peak predictions for the refined jitter sweep, including the two out-of-sample rows, were registered before the sweep completed (`docs/R14-PREREGISTERED-PREDICTIONS.md`, commit 8a627bb) and confirmed in three stages (grid verdict 1d8d7aa; local zoom 4d8310d; pooled Colab replication bd731d5), with the intermediate $k = 4/3$ hypothesis preregistered at 5e51088 and refuted at 1a32758. The campaign freezes at tag `m1-freeze` once the R13 checklist passes outcome-blind self-tests; the companion is frozen at tag `paper1-freeze-v1`. Source-of-record documents for every frozen constant are cited by ruling number in the block comments of the manuscript source; full machinery and proofs are in the supplement (`supplement_m1.tex`). All code, frozen pre-registration documents, per-cell manifests, and evidence bundles are available at [\[repo URL wording — user decision\]](#); funding and acknowledgments are [\[funding/acknowledgments — user decision\]](#).

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